ECON 454 — Business Analytics to Software Implementation

MAXXES

Fan Loyalty & Intelligence Platform

Week 5 Deliverable — Comprehensive Build Document

From Market Research to Build Plan: Mapping Business Analytics into the Live Events Fan Ranking Widget

Session Goal

Students leave with a precise map of how the Maxxes business analytics layer becomes actual software — data objects, dashboard components, XP scoring logic, tier system, recommendation engine, fan segments, market positioning, and Week 5–6 implementation tasks — all aligned to the Master Guiding Document.

01	FRAMING	0 – 10 Minutes

Every visible part of the Maxxes software must trace back to a business analytics question. Maxxes is not a vague fan app. It is a coherent platform with a clear user, a clear business problem, a multi-window data collection layer, an XP-based ranking algorithm, a tier classification system, a recommendation engine, and a reward/action layer that drives revenue for venues and recognition for fans.

The Four Questions Maxxes Answers

Business Question	Data Needed	Analytics Logic	Software Output
Which fans are most valuable to this venue or artist?	Attendance history, spend, social amplification, pre/post engagement	Weighted XP Superfan Score with RFM overlay	Fan Ranking Dashboard
What should high-tier fans access before the general public?	Tier rank (Bronze–Diamond), event proximity, loyalty streak	Tier threshold rules + XP-gated presale eligibility logic	Early Access & Presale Queue
Where are the weak points in fan retention?	Lapsed fans, drop-off events, churn patterns by event type	Lapse detection + 30/60-day recency scoring	Retention Alert Panel
What should venue/artist leadership know about their audience?	Top fan profiles, projected revenue, social reach estimate, segment trends	Aggregated XP value summary with segment breakdown	Business Intelligence Brief

02 DEFINING THE BUSINESS USER & DECISION PROBLEM

10 – 25 Minutes

Maxxes serves two users simultaneously. The fan is the data source and the beneficiary. The venue/artist manager is the analytics consumer and the revenue beneficiary. Both must win for the platform to work.

Primary Users	FANS: Sports fans, music fans, live event attendees seeking XP-based rewards and tier access. BUSINESSES: Venue managers, event producers, artist management, tour marketers, brand sponsors.
Operational Moment	Before the event: fan intent and passion signals captured. During the event: verified presence, in-venue spend, and engagement captured. After the event: social amplification, loyalty behavior, and re-engagement captured.
Business Problem	Fan data is siloed across Ticketmaster, Spotify, Instagram, and venue POS systems. No platform aggregates it into a single ranked fan profile with an auditable XP score. Venues cannot identify their most valuable customers, cannot target premium inventory efficiently, and cannot prove ROI to sponsors. Fans are anonymous despite years of loyal spending.
Bad Outcome if Unsolved	Best seats go to bots and scalpers. Diamond and Platinum fans pay full price for the same access as Bronze attendees. Venues run generic marketing with low conversion. Sponsors pay flat rates with no behavioral proof. Loyal fans churn silently.
Decision Maxxes Enables	FAN: What is the best experience I can access given my XP tier? BUSINESS: Who are my highest-value fans and how do I invest in them to drive more revenue?

Comparable Products by Category

Category	Example Tools	What They Do	Maxxes Differentiator
Ticketing Platforms	Ticketmaster, AXS, SeatGeek	Sell tickets, manage entry, track purchase data	Maxxes adds post-purchase XP-based behavioral ranking Ticketmaster cannot see
Event Discovery	Eventbrite, Songkick, Bandsintown	Surface events by city, genre, artist; notify fans of nearby shows	Maxxes turns discovery into a loyalty pipeline — fans earn XP from attended events
Fan Engagement Apps	Topps Digital, Fanalyze, Rally	Collectibles, fantasy, fan communities	Maxxes ties engagement directly to real-world event access and perks
Loyalty & Rewards	Starbucks Rewards, Delta SkyMiles	Points for purchases, tier-based perks	Maxxes applies the same airline-miles logic to live events with cross-platform signal data
Event Mgmt Platforms	Cvent, Eventbrite, Whova	Registration, check-in, planning workflows	Maxxes focuses on fan value scoring, not event administration
Venue / Ops Software	Momentus, EventPro	Resource planning, space management, event data	Maxxes contributes audience intelligence Momentus cannot generate
Streaming Platforms	Spotify, Apple Music, YouTube	Stream data, listener stats, artist reach	Maxxes connects streaming fandom (XP) to verified real-world attendance

Market Research Share-Out Template

Research Question	Maxxes Answer
What does the product help users decide?	Fans: which experiences match their XP tier. Venues: which fans drive the most value.
What data does it depend on?	Social media activity, ticketing history, in-venue spend, streaming depth, post-event content creation.
What is one feature we can realistically prototype?	A ranked fan leaderboard for a single artist/venue with Bronze–Diamond tier labels and earned perk display.
What feels too big for our capstone?	Full API integrations with Ticketmaster and Spotify. Use mock data for those signals.
What idea can we borrow at prototype scale?	Delta SkyMiles tier logic: earn XP, cross thresholds, unlock perks. Apply to fan attendance and streaming.

Collective Feature List

- Verified attendance tracking (not just ticket purchase) — primary XP multiplier
- Multi-platform data aggregation: social, streaming, ticketing, POS, QR scans
- XP-based fan scoring algorithm with transparent tier logic (Bronze through Diamond)
- Exclusive access rewards tied to tier: presale, VIP, backstage, discounts
- Business-facing Venue Intel dashboard with segment summaries and revenue projection
- Sponsor ROI proof via verified in-venue behavioral data and campaign lift metrics
- Lapsed fan detection and re-engagement targeting with rules-based alerts
- Event discovery feed showing upcoming events in fan's city (Eventbrite/Songkick integration path)
- Vendor and artist partner access to top-ranked fan list for targeted activation

04	MAP THE BUSINESS ANALYTICS COMPONENTS	65 – 80 Minutes

The Maxxes Analytics Stack

Layer	Plain Meaning	Maxxes Implementation
Input Layer	What raw fan data enters the system?	Social posts, ticket purchases, app check-ins, POS transactions, streaming stats, QR scans, post-event content
Normalization Layer	How do we make different signals comparable?	Shared fields: fan_id, event_id, venue_id, signal_type, xp_earned, multiplier, timestamp, source_platform
Descriptive Layer	What is this fan doing right now?	Attendance count, spend per visit, tier status, social reach, streak length, 7/30/90-day activity summary
Scoring Layer	How valuable is this fan?	Weighted XP model: verified attendance (200 XP), watch time (5 XP/hr), listening (3 XP/hr), spend ($1=1 XP), challenges (100–1,500 XP), streak multiplier (1.0x–2.0x)
Prioritization Layer	Which fans earn access first?	Ranked leaderboard sorted by total XP, filtered by event, venue, or artist; tier gates applied to presale windows
Recommendation Layer	What should the fan or venue do?	Fan: You are 340 XP away from Gold tier. Venue: Offer floor upgrade to your top 50 Diamond fans.
Communication Layer	What should leadership know?	Business Intelligence Brief: top fan profiles, XP distribution, projected revenue, lapse risks, campaign lift

The Three Data Windows

Window	Timing	Data Sources	Signals Captured
Pre-Event	Days/hours before doors open	Social media, Ticketmaster API, Spotify listener API, Maxxes platform, Eventbrite/Songkick event feed	Purchase timing, social post frequency, streaming depth, presale engagement, travel distance
Live Event	During the event	Maxxes app check-in, venue POS, in-app QR activations, location services	Verified attendance (+200 XP), arrival time, in-venue spend, activation participation, zone location

Post-Event	Hours/days after the event	Social platforms, Maxxes app, ticketing platforms	Content creation reach, next-show purchase, friend recruitment, review submission, lapse vs. return signal

Governing Rule

The XP scoring model defined here supersedes the 0–100 points scale used in earlier drafts. All team members must use the XP system and Bronze–Diamond tier structure going forward. This section is the single source of truth.

XP Earning Rules

Signal	XP Rule	Reasoning
Verified event attendance	+ 200 XP per event	Direct proof of real-world participation — highest-weight signal
Watch time (game/broadcast)	+ 5 XP per hour	Strong intent signal; tied to true affinity
Listening time (Spotify/music)	+ 3 XP per hour	Useful signal; more passive than watching
In-venue / merch spend	+ 1 XP per $1 spent	Important but should not dominate the model alone
Challenge completion	+ 100 to + 1,500 XP	Drives repeat engagement and visible progression
Streak modifier	1.0x – 2.0x on selected base XP	Rewards consistency over isolated activity spikes
Post-event social amplification	+ 10 – 50 XP per post (reach-adjusted)	Measures fan-generated promotion and community building
Friend recruitment	+ 75 XP per converted referral	Organic growth signal with high-quality downstream value

RFM Overlay

The XP model incorporates Recency, Frequency, and Monetary value logic. A fan who spent heavily two years ago should not automatically outrank a fan with strong recent attendance. Recency logic — scoring activity windows at 7, 30, and 90 days — keeps the ranking current and actionable.

- **Recency:** XP from the past 30 days weighted 2x over XP older than 90 days.
- **Frequency:** Streak multiplier (up to 2.0x) rewards fans who engage consistently rather than sporadically.
- **Monetary:** Spend XP (1 XP/$1) contributes to the model but is capped at 20% of total score weight to prevent wealth-only rankings.

Tier System — Bronze through Diamond

The following tier structure is the official system for Maxxes. All UI components, presale logic, perk eligibility, and operator dashboard filters must reference these exact tier names and XP thresholds.

Tier	XP Threshold	Interpretive Meaning
Bronze	0 – 2,499	Entry-level or casual engagement
Silver	2,500 – 4,999	Consistent but still developing loyalty
Gold	5,000 – 9,999	Meaningful repeat engagement; practical rewards relevance
Platinum	10,000 – 24,999	High-value fan across multiple signal types
Diamond	25,000+	Top-tier fan; scarce access eligibility

Priority of Dashboard Surfaces

Per the Master Guiding Document, the build priority order is: (1) Venue Intel Dashboard — HIGHEST, (2) Fan Home / My Journey screen — HIGH, (3) Rewards Marketplace — HIGH, (4) Challenges — MEDIUM, (5) Partner / Activation View — MEDIUM. If scope must be cut, preserve the Venue Intel Dashboard and Fan Home screen above all others.

06 CANONICAL DATA OBJECTS

Single Source of Truth

These 9 canonical entities are the definitive data model for Maxxes. They supersede the 12-object list in earlier drafts. Ethan King (Back End) owns the schema; all other roles must reference these objects when naming fields in their components.

Entity	Example Fields	Purpose
Fan	fan_id, city, preferred_team_artist, current_tier, xp_total, streak_days	Master user record and ranking anchor
FanEvent	event_type, source_platform, duration_minutes, amount_usd, venue_id, campaign_id, timestamp	Immutable behavioral signal log — every XP-earning action recorded here
XPTransaction	source_event_id, xp_earned, multiplier, xp_reason, calculated_at	Audit trail for scoring — makes the model explainable and testable
Challenge	challenge_type, required_action, xp_reward, sponsor_id, active_window	Defines gamified actions fans can complete for XP
ChallengeProgress	fan_id, challenge_id, current_progress, target_progress, completed_at	Tracks per-fan challenge state and completion status
Reward	tier_required, xp_cost, quantity_available, reward_category, partner_id	Defines redeemable experiences and their eligibility requirements
RewardRedemption	reward_id, fan_id, xp_spent, status, redeemed_at	Tracks reward usage and remaining catalog inventory
Campaign	campaign_type, audience_segment, offer_type, channel, start_date, end_date	Supports attribution logic and campaign lift measurement
VenueSummary	venue_id, avg_xp, tier_distribution, presale_conversion_rate, churn_flag_count, generated_at	Operator-facing aggregate snapshot powering the Business Intel Dashboard

Required Dashboard Components — Priority Order

#	Component	Serves	Build Priority
1	Venue Intel Dashboard	Operator	HIGHEST — build first
2	Fan Home / My Journey	Fan	HIGH — build second
3	Rewards Marketplace	Fan	HIGH — build third
4	Superfan Score Card	Fan	HIGH — embedded in Fan Home
5	Fan Leaderboard	Fan + Venue	HIGH — embedded in both views
6	Next Milestone Tracker	Fan	Medium — part of Fan Home

7	Challenges Panel	Fan	Medium — standalone screen
8	Lapse Risk Alert	Operator	Medium — embedded in Venue Intel
9	Revenue Projection Panel	Operator	Medium — embedded in Venue Intel
10	Partner / Activation View	Operator	Medium — build if time permits
11	Data / Assumptions Panel	All	Low — add at end for transparency

07	TRANSLATE ANALYTICS INTO SOFTWARE REQUIREMENTS	80 – 100 Minutes

Example 1 — Superfan XP Score Card

Business Question	How loyal and valuable is this fan compared to all other fans of this artist/venue?
Data Fields Needed	fan_id, xp_total, current_tier, streak_days, xp_last_30_days, xp_last_90_days — all from Fan and XPTransaction entities
Calculation / Rule	Sum all XPTransaction.xp_earned for this fan. Apply streak multiplier. Apply recency weighting (30-day XP × 2.0). Classify tier using Bronze–Diamond thresholds. Display next milestone XP gap.
UI Component	Superfan Score Card — circular XP gauge, tier badge (color-coded Bronze–Diamond), points breakdown by signal type, next milestone indicator, streak counter
Owner Roles	Ethan: scoring weights and XPTransaction schema. Luke: renders card and gauge. Kayah: validates data contract. Justin: tests tier threshold logic. Kyle: badge visual hierarchy.
Week 5 Task	Finalize XP weights, confirm tier XP thresholds, build mock Fan and XPTransaction JSON dataset (25+ fans, 90 days of history).
Week 6 Task	Implement live XP calculation from mock data; render score card with correct tier logic and milestone display.

Example 2 — Fan Leaderboard (Ranked Access Queue)

Business Question	Which fans have earned priority access to presales, VIP upgrades, and exclusive experiences?
Data Fields Needed	fan_id, fan_name, xp_total, current_tier, events_attended, artist_id, venue_id, perk_eligibility_list, last_event_date, streak_days — from Fan and RewardRedemption entities
Calculation / Rule	Sort all fans by xp_total descending. Apply tier bands: Diamond (25,000+), Platinum (10,000–24,999), Gold (5,000–9,999), Silver (2,500–4,999), Bronze (0–2,499). Top-tier fans receive first access to presale windows and premium upgrade offers.
UI Component	Ranked leaderboard table with tier badges, XP score, attendance count, streak status, and perk unlock indicators
Owner Roles	Ethan: tier band thresholds and sort logic. Luke: builds sortable table with tier filter. Justin: validates sort and tier accuracy. Kyle: badge and tier color system.
Week 5 Task	Define tier XP thresholds; design leaderboard layout; build mock fan roster (25+ profiles across all 5 tiers).

Week 6 Task	Render leaderboard from mock data; implement tier filter; show perk unlock states by tier.

Example 3 — Venue Intelligence Brief (Operator View)

Business Question	Who are my top fans, what are they worth, and what should I offer them?
Data Fields Needed	VenueSummary fields: top_fans_snapshot (top 100 by XP), projected_revenue, social_reach_estimate, lapsed_fans_count, avg_spend_elite_vs_general, tier_distribution, next_event_id — from VenueSummary and Fan entities
Calculation / Rule	If a Gold+ fan has not generated a FanEvent in 60+ days, flag as lapse risk. Projected revenue = avg_spend_per_visit × superfan_count × events_remaining. Social reach = sum of follower counts for fans who posted post-event content.
UI Component	Business Brief panel: top fan tiles ranked by XP, tier distribution bar chart, revenue projection, lapse risk alerts with recommended action, segment breakdown cards
Owner Roles	Ethan: lapse and projection rules, VenueSummary mock data. Luke: renders brief panel and chart. Kyle: executive-readable layout. Justin: checks revenue math and lapse flag logic.
Week 5 Task	Draft brief rules and VenueSummary data contract; design brief layout; define projection formula.
Week 6 Task	Generate brief from mock fan roster; render lapse flags; display revenue estimate and tier distribution.

08	RECOMMENDATION ENGINE — WEEK 5 ADDITION	Reference

Why This Section Exists

The recommendation engine converts descriptive analytics into interventions. This is one of the most important differentiators for Maxxes as a business analytics platform, not just a fan app. Each recommendation card has a trigger condition, a target audience, a recommended action, a business rationale, and a prototype output. The engine is rules-based — transparent, testable, and appropriate for a five-week capstone.

Recommendation Rules — Full Engine

Use Case	Trigger Condition	Recommended Action	Business Rationale	Prototype Output
New fan activation	Moderate early XP but no ticket purchase or attendance within 30 days	Offer low-barrier challenge and first-ticket incentive	Early conversion improves long-term retention rate	Activate New Fans card
At-risk fan rescue	Gold, Platinum, or Diamond fan XP declines materially over 30–60 days	Trigger re-engagement campaign or exclusive perk offer	Retention is cheaper than reacquisition for high-value fans	At-Risk Gold+ Fans alert
Presale prioritization	Upcoming high-demand event with constrained premium inventory	Apply XP threshold or tier gate (Gold+ minimum) for early access window	Makes presale distribution defensible and targeted rather than first-come	Presale Access Rule panel
Reward optimization	High-visibility reward category with low redemption rate	Retune XP cost, improve reward description, or replace with higher-value perk	Rewards should reinforce behavior, not decorate the interface	Reward Performance table
Campaign reinforcement	Strong XP lift in one segment; weak or negative lift in another	Scale where it works; revise or discontinue where it does not	Improves marketing efficiency and reduces wasted spend	Campaign Lift by Segment view
VIP governance	Scarce premium experiences require fairer allocation	Sort eligible fans by XP plus recency of attendance (last 60 days)	Makes access earned rather than arbitrary or wealth-dependent	VIP Candidate List card
Partner targeting	Sponsor asks which fan segments respond best to their category	Recommend partner-fit segments based on spend behavior and engagement signal overlap	Improves sponsorship credibility and enables performance-based deals	Best Partner-Fit Segment card

Event-day upsell timing	Venue wants to improve in-event spend conversion	Use FanEvent timing patterns to place offers at highest-response windows	Timing often matters as much as the offer itself	Best Activation Window recommendation

Dashboard Layout for Recommendation Cards

Dashboard Zone	Recommended Contents	Reason
Top row (KPIs)	Active fans, At-risk fans, Presale conversion rate, Reward redemption rate	Fast health check — operator sees system status in under 10 seconds
Middle row (Descriptives)	Tier distribution chart, Campaign lift by segment, XP and spend trend panel	Core descriptive analytics — answers what is happening and why
Bottom row (Actions)	Recommendation action queue, Partner activation performance, Data/Assumptions status box	Turns analytics into operational decisions — the recommendation engine output lives here

Rule for Every Chart or Card

Every dashboard element must answer at least one of three questions: What happened? Why does it matter? What should we do next? If a component answers none of these, it should be removed or redesigned.

09 **FAN SEGMENTS — WEEK 5 ADDITION** Reference

Why Segments Matter

Segments are the analytical bridge between individual fan scores and operator decisions. A venue manager cannot act on 50,000 individual fan profiles, but they can act on 8 clearly defined segments. Every operator dashboard recommendation should reference a segment, not just an individual fan.

Canonical Seed Segments — Maxxes

Segment	Definition	Typical XP Profile	Operator Action
New Fans	Joined within last 30 days; first 1–2 FanEvents logged	0 – 500 XP	Offer first-attendance incentive; low-barrier challenge to build habit
Casual Fans	Active but infrequent; attend 1–2 events per season; low streak	500 – 2,499 XP (Bronze)	Engagement campaigns to increase frequency; streak-starter challenges
Core Fans	Consistent attendance and streaming; 3+ events per season; active streak	2,500 – 9,999 XP (Silver–Gold)	Loyalty rewards, presale access at Gold; social amplification challenges
Superfans	High XP across all signal types; top 15% of fan base; strong streak	10,000 – 24,999 XP (Platinum)	VIP experiences, partner perks, sponsor activations; brand ambassador targeting
Diamond Fans	Top 5% by XP; multi-year loyalty; highest spend and attendance	25,000+ XP (Diamond)	Backstage access, artist/team meet-and-greets, exclusive pre-release content
At-Risk High-Value Fans	Gold, Platinum, or Diamond tier but no FanEvent in 30–60 days; XP declining	Previously high XP, recent activity near zero	Re-engagement offer, personalized perk, lapse flag alert in Venue Intel
Promo-Responsive Fans	Strong XP lift correlated with past QR or campaign activations	Variable XP; high campaign_id FanEvent density	Target with QR challenges, social share campaigns, sponsor activation offers
High-Spend / Low-Attendance	High spend XP from POS but low verified-attendance XP	High monetary signal; low attendance signal	Incentivize event attendance; offer experience upgrade tied to first check-in
High-Attendance / Low-Spend	Frequent verified attendance but low in-venue spend	High attendance XP; low spend XP	Targeted concession or merch offer; spend challenge with XP reward

How Segments Power the Dashboard

- **Tier distribution chart:** Shows how many fans sit in each segment — the core health metric for any venue.
- **Segment growth/decline:** Is the Core Fan segment growing or shrinking month over month? This drives programming and marketing decisions.
- **Campaign lift by segment:** Which segment responded most to the last QR campaign? Informs where to concentrate future spend.
- **Churn signal count:** How many At-Risk High-Value fans are flagged this week? Triggers the re-engagement recommendation card.
- **Average XP and spend by segment:** Identifies which segments drive the most revenue — essential for sponsor pitch conversations.

Week 5 — Design the MVP

Output	Maxxes-Specific Meaning	Evidence for Lab Journal
Feature list	What Maxxes will and will not include at MVP. Must include: XP score card, leaderboard (Bronze–Diamond), perk panel, venue intel brief, recommendation cards. Exclude: live API integrations — mock data only.	Written feature scope document with in/out decisions justified
Dashboard wireframes	One-screen fan-facing view (My Journey) and one-screen operator view (Venue Intel). Fan view shows XP score, tier badge, leaderboard rank, streak, and perk status. Operator view shows segment distribution, XP trend, campaign lift, lapse alerts, and recommendation queue.	Sketches or Figma frames for both views
Data dictionary	All 9 canonical entities defined with field names, types, and descriptions. Mock JSON examples for Fan, FanEvent, XPTransaction, VenueSummary, and Challenge.	Completed data dictionary document
Analytics map	Business question → data field → calculation rule → UI component. One row per dashboard component.	Completed analytics map table (use Section 07 as template)
Role task plan	Who owns XP algorithm logic, frontend rendering, data schema, QA, and design for each component. Reference the canonical entities, not improvised field names.	Role assignment matrix aligned to Section 07 examples
Early design assets	Tier badge designs (Bronze–Diamond color system), XP score card mockup, color palette, segment card layout.	Screenshots of design assets in lab journal

Week 6 — Build the Core

Output	Maxxes-Specific Meaning	Evidence for Lab Journal
MVP Alpha	First working prototype with mock data powering all required components. XP score renders. Leaderboard sorts by XP with correct tier badges. Perk panel reflects tier. Business brief generates from fan roster. Recommendation cards fire based on trigger rules.	Screenshots of working prototype

Integrated screen flow	Fan view and operator view render coherent data from the same mock fan roster. Tier logic is consistent across both views. Segment labels match canonical definitions.	Screen recording or screenshot sequence
Logic notes	XP weights documented. Tier XP thresholds documented. Lapse rule (60-day threshold) documented. Revenue projection formula documented. RFM recency weighting documented.	Logic notes file in lab journal
Screenshot evidence	At minimum: XP score card rendering with tier badge, leaderboard sorted by XP, perk panel with at least 2 tier-gated unlock states, business brief with top fan tiles and lapse flag, one recommendation card firing from mock data.	4+ labeled screenshots
Lab journal entry	Each team member: role-specific reflection, what they built, what decisions they made, what changed, what they would improve next week.	Individual written entries

11	RASPBERRY PI & SMALL LANGUAGE MODEL TIE-IN	Week 5 Addition

Why This Component Exists

The Raspberry Pi-hosted small language model device is a required student outcome per the Master Guiding Document. It is not the capstone's central claim — Maxxes is. But it is a strong interview talking point, a practical demonstration of applied AI experimentation, and a useful tool for supporting the prototype ecosystem throughout the build.

What the Raspberry Pi Device Does in the Maxxes Context

Use Case	What It Does	Who Benefits
Local project explainer chatbot	A small language model runs locally on the Pi and answers questions about how Maxxes works — how XP is calculated, what the tier thresholds are, what a lapse flag means	Investors, interviewers, class visitors who want to ask questions without a team member present
Scoring model explainer	The SLM can walk through the XP scoring logic step by step when given a fan profile — useful for demo walkthroughs and portfolio presentations	Employers; demo audiences
Requirements summarizer	Paste in a Week 5 or Week 6 lab journal prompt and the SLM summarizes key decisions, blockers, and next steps in professional language	All team members for lab journal drafting
Mock analyst assistant	The SLM can respond to operator-style questions like 'Which of my fan segments is most at risk this week?' using the seeded dataset and rules-based logic	Kyle and Luke for demo scripting; Justin for QA narrative
Lab journal drafting helper	Team members can use the local SLM to draft role-specific lab journal reflections, then edit for personal voice and accuracy	All team members
Portfolio demonstration artifact	The device itself is a physical artifact that proves hands-on AI experimentation — photographed or shown in portfolio website	All team members for personal websites

Implementation Constraints

- The Raspberry Pi device supports the Maxxes ecosystem — it does not replace the core analytics or UI work.

- The SLM runs locally: no cloud dependency, no API key required, appropriate for a classroom and interview setting.
- Any AI or SLM layer should support explanation, documentation, or lightweight assistance rather than substitute for the core scoring logic built by Ethan.
- The class should state clearly in the final presentation that the Pi demonstrates practical AI experimentation — not production AI deployment.
- Target SLM options: LLaMA 3 (8B), Mistral 7B, or Phi-2 — all runnable on Raspberry Pi 5 with quantized weights.

Raspberry Pi Tie-In by Role

Role	Pi Contribution	Portfolio Talking Point
Kayah Whipps — Technical Lead	Configure Pi hardware, install SLM runtime (llama.cpp or Ollama), define prompt templates for Maxxes use cases	Deployed a locally hosted small language model on embedded hardware to support a business analytics prototype
Kyle Rumble — Design / Creative	Design the demo flow that incorporates the Pi — how it appears in the live demo, what the screen shows, how it supports the narrative	Integrated a physical AI device into a product demo as a UX and storytelling element
Luke Pollard — Front End	Optionally build a simple web UI that sends queries to the Pi's local SLM endpoint — a lightweight chat interface for the explainer use case	Built a minimal frontend interface connecting to a locally hosted language model
Ethan King — Back End	Write the system prompt that gives the SLM accurate knowledge of the Maxxes data model, XP rules, tier thresholds, and segment definitions	Engineered structured prompts for a domain-specific local language model aligned to a business analytics data model
Justin Hurst — QA & Docs	Test the SLM's accuracy on Maxxes-specific questions; document where it performs well and where it needs prompt refinement; include in final documentation	Conducted structured QA on a local SLM deployment and documented performance against business logic requirements

12 EXIT TICKET / LAB JOURNAL SETUP

112 – 120 Minutes

Each team member submits the following individually before leaving the session. This becomes the opening entry in the Week 5 lab journal and the first line of evidence for each student's role-specific portfolio documentation.

Field	Maxxes Example Answer
Name	(Your name)
Role	Algorithm Owner / Frontend Owner / Data Owner / QA / Design Owner
Analytics component I am responsible for	e.g., XP scoring model and Bronze–Diamond tier threshold logic
Canonical data entities my component uses	e.g., Fan, XPTransaction, FanEvent — field names must match Section 06
UI output my component affects	e.g., Superfan XP Score Card, Tier Badge, Leaderboard rank position, Next Milestone Tracker
My Week 5 deliverable	e.g., Finalize XP weights document and build mock Fan/XPTransaction JSON dataset (25+ profiles, 90 days history)
My Week 6 deliverable	e.g., Implement XP scoring algorithm; render score card from mock data with correct tier logic and milestone display
One market research insight that changed how I understand the widget	e.g., Delta SkyMiles proves fans will change behavior for tier status — Maxxes applies the same psychology to live events and streaming
One blocker or question I need resolved	e.g., How do we weight social reach vs. attendance without live Spotify API access in the prototype?
Raspberry Pi use case I will support	e.g., Writing the system prompt that gives the SLM accurate XP rules and tier definitions for the explainer chatbot

13	MAXXES WIDGET BUILD MAP	Instructor Reference

Business Problem	Fan data is siloed across ticketing, social, and streaming platforms. Venues cannot identify their most valuable customers. Fans are anonymous despite years of loyal spending and cannot access experiences that match their true devotion.
Primary Users	FANS: seeking XP rewards, tier recognition, and priority access. VENUES / ARTISTS / MANAGERS: seeking audience intelligence to drive revenue and retention.
Core Decision	FAN: What have I earned and what can I access next? VENUE: Who are my best fans and how do I invest in them to grow revenue?
Market Positioning	Maxxes sits between the venue and the fan as the engagement and intelligence layer. It does not replace Ticketmaster or Spotify. It aggregates what they cannot see — cross-platform behavioral history tied to a verified, attending human — and turns that into a ranked fan profile every party in the ecosystem will pay to access.
Tier System	Bronze (0–2,499 XP) → Silver (2,500–4,999) → Gold (5,000–9,999) → Platinum (10,000–24,999) → Diamond (25,000+)
Scoring Model	XP-based: attendance +200/event, watch time +5 XP/hr, listening +3 XP/hr, spend +1 XP/$1, challenges +100–1,500, streak multiplier 1.0x–2.0x, with RFM recency weighting
Week 5 Goal	Design the MVP: define the feature set, wireframe both views, build the data dictionary using canonical entities, complete the analytics map, assign roles, produce early design assets, and confirm the Raspberry Pi use case for each team member.
Week 6 Goal	Build the core: first working prototype using mock data, all required components rendering with correct XP and tier logic, recommendation engine firing from trigger rules, screenshot and lab journal evidence documented.

Core Analytics Logic Summary

- **Descriptive metrics:** What is this fan doing? Attendance count, spend per visit, social posts, XP by signal type, streak length, 7/30/90-day activity summary.
- **XP Superfan Score:** How valuable is this fan? Weighted multi-signal XP model with RFM recency overlay.
- **Tier classification:** What level does this fan belong to? Bronze, Silver, Gold, Platinum, Diamond — based on cumulative XP.
- **Prioritization:** Which fans earn access first? Sorted leaderboard by XP, filtered by artist, venue, or event.
- **Recommendation engine:** What should the fan or venue do next? Eight rules-based cards covering activation, rescue, presale, rewards, campaigns, VIP, partners, and event-day upsell.

- **Segment analysis:** How is the audience distributed? Nine canonical segments from New Fans to Diamond — each with a defined operator action.
- **Business intelligence summary:** What does leadership need to know? Top fan profiles, XP distribution, revenue projection, lapse risks, campaign lift by segment.

Closing Framework

The point of this lab is to make the Maxxes build defensible. Every dashboard component should trace back to a business question. Every business question should trace back to data. Every data field should support an XP calculation, a tier rule, a segment definition, or a recommendation trigger. If students can explain that chain — from a fan's third concert attendance to a Gold tier upgrade to a backstage pass to a revenue projection in the venue dashboard — they are not just building an app. They are building a business analytics platform with a real market, real users, and a defensible data moat.